

company has announced plans to start shipping EUV light sources this year.

It is expected that by 2012, production systems for volume manufacturing using EUV lithography will be available. However, this is assuming that certain hurdles are mounted. Making masks of sufficiently high quality, for instance, is still a challenge. Another is checking for defects in those masks initially and after use. Companies such as KLA-Tencor Corporation and Carl Zeiss SMT are making mask inspection tools capable of handling 22nm logic node EUV masks. However, these work using 193nm sources, and it is believed that they will work only up to 16nm process technology. So, what after that?

A review of the recently held *SPIE Advanced Lithography - EUVL Conference* by Vivek Bakshi (<http://semind.com/blog/2011/03/07/2011-spie-advanced-lithography-%E2%80%93-euvl-conference-review/>) provides an excellent overview of recent methods in lithography and metrology.

Experiments with newer materials

Silicon has dominated the semiconductor manufacturing space for ages and, frankly, it is still one of the best materials for chips. However, new materials are being experimented for newer applications and smaller geometries, but none has moved beyond research to volume production yet. As long as we continue with the same material, improvements will only be incremental.

Electronic devices such as touchscreens, flexible displays, printable electronics, solidstate lighting and thin-film photovoltaics, for example, would be much better off with flexible transparent conductors. Standard indium-tin-oxide films are unlikely to satisfy future needs due to losses in conductivity on bending and the escalating cost of indium which is in limited supply.

"Recent research related to graphene indicates that it may be suitable as a transparent conductor. Graphene hybrids with carbon nanotubes, for

example, may prove to be especially interesting," explains Prof. Jadcherla.

Some companies have also been experimenting with polymer-based semiconductor manufacturing. Other materials like bismuth telluride and molybdenite have also been checked up. However, carbon nanotubes show much more promise than any other option, although not in the short-term. A lot of disparate but promising research is happening on that front. Last year, IBM revealed carbon nanotube air filters for semiconductor manufacturing. In January, TSMC patented a carbon nanotube bond pad structure and the methods for using it. In February, Prof. Yutaka Ohno from Nagoya University in Japan and Prof. Esko Kauppinen from Aalto University

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in Finland shared details of a simple and fast process to manufacture high-quality carbon nanotube-based thin-film transistors on a plastic substrate. This can go a long way in producing cheap, powerful electronics. However, it might be years before carbon nanotubes really come into semiconductor manufacturing in a big way.

"Nanotubes and other materials are not yet on the horizon as far as volume manufacturing is concerned. This trend is unlikely to reverse in the next 5-10 years. But yes, a revolution is expected. Carbon nanotubes are expected to become mainstream in a decade," says Sathe.

Changes galore

There are lot more trends catching up in the space. For example, device makers are moving towards a fab-lite

model, whereby they handle only the designing and outsource the manufacturing to foundries, in order to avoid the cost of setting up a fab and the problems caused by changes in technology and demand.

Another trend is the focus on reducing the time-to-market. Since not much can be done to reduce the actual time of manufacture, the onus has shifted to the design stage—more capable tools and processes are helping to reduce the time taken at this stage.

Sathe agrees, "The time taken for manufacturing and post-manufacturing process is generally fixed. Hence the emphasis is on reducing the design cycle time. An approach preferred these days in creating bigger system-on-chips is to reuse or buy intellectual property, and to work with design partners rather than in-house development. The integrated device manufacturers are looking at a faster time-to-market and increasing market share rather than a complete in-house design approach. New methodologies brought in by electronic design automation vendors, e.g., Magma's methodology to accelerate analogue porting and optimisation, also help in this."

There is also an attempt to move from the current 30.5cm (12-inch) wafers to 45.7cm wafers. The 30.5cm, 300mm wafer-based technology, which itself is kind of new, yields 2.25 times more chips per wafer than the older 20.3cm (8-inch), 200mm wafers, yet they take just about the same time to pass through a factory, reducing the cost per chip and significantly boosting total monthly output. A 45.7cm wafer plant would show a similar reduction in per-chip cost and increase in output. However, as of now, nobody can afford a 45.7cm fab. An industry conglomeration, including TSMC, is looking at how to make 45.7cm fabrication more affordable. The company expects to bring it into operation by 2015. Let us wait and see what other changes happen by then. ●

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